

Assignment 1

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Questions

1. Based on the data above, is the assumption of zero-mean Gaussian noise reasonable? What values of the variances σ_q^2 and σ_r^2 should we use?

$$w_k \sim \mathcal{N}(0, \sigma_q^2), \quad n_k \sim \mathcal{N}(0, \sigma_r^2) \quad (1.2)$$

Solution:

The error histograms in the dataset show that both the odometry and the range measurements are centered around zero and have shapes that closely resemble Gaussian distributions. It is therefore reasonable to model both noise sources as zero-mean Gaussian.

From Fig. 1.6, the standard deviation of the range measurement error is

$$\text{StdDev}_r = 0.019155 \text{ m},$$

which gives the measurement noise variance

$$\sigma_r^2 = (0.019155)^2 \approx 3.6692 \times 10^{-4}.$$

The odometry error has a standard deviation of

$$\text{StdDev}_v = 0.047554 \text{ m/s}.$$

Since the process noise affects the position through the sampling period $T = 0.1$ s, the resulting position noise standard deviation is

$$\sigma_q = T \cdot \text{StdDev}_v = 0.1 \times 0.047554.$$

Squaring this value gives the process noise variance:

$$\sigma_q^2 = (0.1 \cdot 0.047554)^2 \approx 0.0023.$$

These variances reflect the observed spread of the sensor errors and are used in the batch linear-Gaussian estimator.

2. Write out an expression for the batch linear-Gaussian objective function that we will seek to minimize:

$$J(x_{1:K} \mid u_{1:K}, y_{1:K}), \quad (1.3)$$

where K is the index of the maximum time.

Solution: we estimate the full state trajectory $x_{1:K}$ by combining the motion model and the measurement model in a single batch formulation. The system dynamics and observations are

$$x_k = x_{k-1} + Tu_k + w_k, \quad y_k = x_k + n_k,$$

where $w_k \sim \mathcal{N}(0, \sigma_q^2)$ and $n_k \sim \mathcal{N}(0, \sigma_r^2)$.

According to the batch solution (using MAP), form those matrices as:

$$\mathbf{z} = [\tilde{x}_0, Tu_1, \dots, Tu_k \mid y_0, \dots, y_k]^T$$

$$\mathbf{x} = [x_0, x_1, \dots, x_k]^T$$

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ -A_0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -A_1 & 1 & 0 & 0 & 0 \\ 0 & 0 & \ddots & 1 & 0 & 0 \\ 0 & 0 & 0 & -A_{K-1} & 1 & 0 \\ C_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & C_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & C_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & 0 & C_K & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & \ddots & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$W = \begin{bmatrix} 0.01 & 0 & 0 & \dots & 0 & 0 & 0 & 0 \\ 0 & \sigma_q^2 & 0 & \dots & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_q^2 & \dots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & \sigma_q^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & \sigma_r^2 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & \sigma_r^2 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & \sigma_r^2 \end{bmatrix}$$

Under the assumption of independent Gaussian noise, the resulting batch cost function is the standard weighted least-squares objective:

$$J(x_{1:K}) = \frac{1}{2} (\mathbf{z} - H\mathbf{x})^T W^{-1} (\mathbf{z} - H\mathbf{x}).$$

This function is minimized to obtain the MAP estimate of the trajectory.

3. Derive an expression for the optimal position estimates,

$$x_{1:K}^* = \arg \min_x J(x_{1:K} \mid u_{1:K}, y_{1:K}) \quad (1.4)$$

using the method of least squares.

Solution:

To obtain the optimal trajectory estimate $x_{1:K}^*$, we minimize the batch cost function

$$J(x_{1:K}) = \frac{1}{2} (\mathbf{z} - H\mathbf{x})^T W^{-1} (\mathbf{z} - H\mathbf{x}),$$

where H , $\mathbf{z}$, and W were defined previously. Since this cost function is a quadratic form in $\mathbf{x}$, its minimum can be found by setting its gradient to zero. Taking the derivative of $J(\mathbf{x})$ gives

$$\frac{\partial J}{\partial \mathbf{x}} = -H^T W^{-1} (\mathbf{z} - H\mathbf{x}).$$

Setting this expression equal to zero leads to the normal equations

$$H^T W^{-1} H \hat{\mathbf{x}} = H^T W^{-1} \mathbf{z}.$$

Solving this linear system yields the MAP estimate of the full trajectory:

$$\hat{\mathbf{x}} = (H^T W^{-1} H)^{-1} H^T W^{-1} \mathbf{z},$$

which corresponds to the optimal batch estimate

$$x_{1:K}^* = \hat{\mathbf{x}}.$$

Because the matrix $H^T W^{-1} H$ is symmetric and positive definite, the system can be efficiently solved using Cholesky factorization or related numerical techniques.

4. This dataset has $K = 12709$. It may be computationally intractable to solve for all K positions using a batch method. Suppose we only want to solve for a subset of these poses,

$$x_\delta, x_{2\delta}, x_{3\delta}, \dots, x_K, \tag{1.5}$$

where $\delta \geq 1$.

We will only use the laser rangefinder reading at the corresponding timesteps,

$$r_\delta, r_{2\delta}, r_{3\delta}, \dots, r_K, \tag{1.6}$$

but want to use all of the odometry speed measurements.

Modify the objective function and expression for the optimal position estimates to handle this case.

Solution:

To reduce computation, we estimate only every δ -th pose,

$$x_0, x_\delta, x_{2\delta}, \dots, x_K, \quad \delta \geq 1,$$

and keep only the corresponding range measurements

$$y_\delta, y_{2\delta}, \dots, y_K.$$

All odometry measurements remain fully used.

Reduced State Vector. Let

$$n = \left\lfloor \frac{K}{\delta} \right\rfloor,$$

and define the reduced state as

$$\mathbf{x}_\delta = [x_0, x_\delta, x_{2\delta}, \dots, x_{n\delta}, x_K]^T.$$

Modified Measurement Vector. Odometry over each δ -step interval is accumulated,

$$\Delta x_j = \sum_{i=(j-1)\delta+1}^{j\delta} T u_i,$$

yielding

$$\mathbf{z}_\delta = [\tilde{x}_0, \Delta x_1, \Delta x_2, \dots, \Delta x_n \mid y_\delta, y_{2\delta}, \dots, y_K]^T.$$

The matrices become:

$$H_\delta = \left[\begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 0 \\ -A_0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -A_\sigma & 1 & 0 & 0 & 0 \\ 0 & 0 & \ddots & 1 & 0 & 0 \\ 0 & 0 & 0 & -A_{n\sigma} & 1 & 0 \\ \hline C_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & C_\sigma & 0 & 0 & 0 & 0 \\ 0 & 0 & C_{2\sigma} & 0 & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & 0 & C_K & 0 \end{array} \right] = \left[\begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & \ddots & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ \hline 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right]_{2(n+2) \times (n+2)}$$

$$W_\delta = \left[\begin{array}{ccccccccc} \tilde{P}_0 & 0 & 0 & \dots & 0 & 0 & 0 & 0 \\ 0 & Q_\sigma & 0 & \dots & 0 & 0 & 0 & 0 \\ 0 & 0 & Q_{2\sigma} & \dots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & Q_K & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & R_0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & R_\sigma & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & R_K \end{array} \right]$$

$$= \left[\begin{array}{ccccccccc} 0.01 & 0 & 0 & \dots & 0 & 0 & 0 & 0 \\ 0 & \sigma_q^2 & 0 & \dots & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_q^2 & \dots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & \sigma_q^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & \sigma_r^2 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & \sigma_r^2 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & \sigma_r^2 \end{array} \right]_{2(n+2) \times (n+2)}$$

Here we need to point out that if $\delta = 1$, the dimension changes slightly because $\frac{K}{\delta}$ is exact. Thus all $(n + 2)$ should be changed to $(n + 1)$.

Therefore:

$$\mathbf{x}_\delta^* = (H_\delta^T W_\delta^{-1} H_\delta)^{-1} H_\delta^T W_\delta^{-1} \mathbf{z}_\delta \quad (16)$$

5. Write a Matlab or Octave script to solve for the robot positions for

$$\delta = 1000, 100, 10, 1. \quad (1.7)$$

Make the following figures and write comments about your findings. The true robot positions $x_{k\delta}$ are available in the `x_true` array.

(a) Plot on the same axes:

- the error $x_{k\delta}^* - x_{k\delta}$ vs. $t_{k\delta}$ (solid line),
- the uncertainty envelope $\pm 3\sigma_{x_{k\delta}}$ vs. $t_{k\delta}$ (dotted lines).

The variance $\sigma_{x_{k\delta}}^2$ can be extracted from the least-squares solution. Make one plot for each δ .

(b) Make a histogram of the errors $x_{k\delta}^* - x_{k\delta}$ for each value of δ . Discuss how this aligns with the predicted uncertainty from the least-squares formulation.

Solution:

The batch estimator was evaluated using four downsampling intervals, $\delta \in \{1000, 100, 10, 1\}$. The results show a clear and monotonic relationship between δ , the estimation accuracy, and the model uncertainty.

Error Over Time: For $\delta = 1000$, the estimation error is relatively large because the system receives only a few laser updates and the estimate is dominated by odometry drift. As δ decreases, laser measurements occur more frequently and compensate for odometry accumulation. For $\delta = 1$, the error is tightly bounded and remains close to zero.

Trajectory Comparison: When δ is large, the estimated trajectory noticeably deviates from the ground truth. With smaller δ , the estimate follows the true trajectory more closely, and for $\delta = 1$ the two curves nearly overlap.

Uncertainty Envelope ($\pm 3\sigma$): The predicted covariance decreases as δ becomes smaller. For $\delta = 1000$, the uncertainty band is wide, indicating limited information from sparse measurements. As δ decreases, the band becomes narrower, showing that more measurement updates reduce uncertainty. The results for $\delta = 1$ exhibit the smallest variance.

Error Histogram and Gaussian Fit: For large δ , the error histogram is wide and irregular, reflecting limited measurement constraints. As δ decreases, the histogram becomes narrower and more Gaussian-shaped. The distribution for $\delta = 1$ has the smallest variance and matches the Gaussian fit closely.

Conclusion A smaller δ provides more frequent measurement updates, leading to:

- lower estimation error,

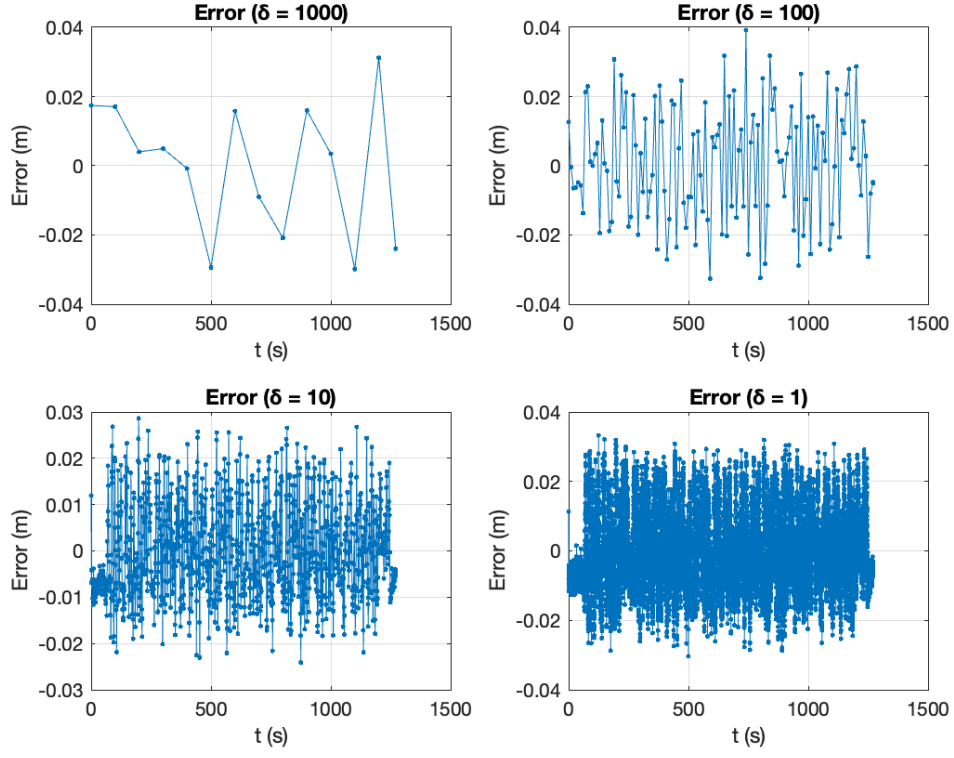


Figure 1: Estimation error for different values of δ .

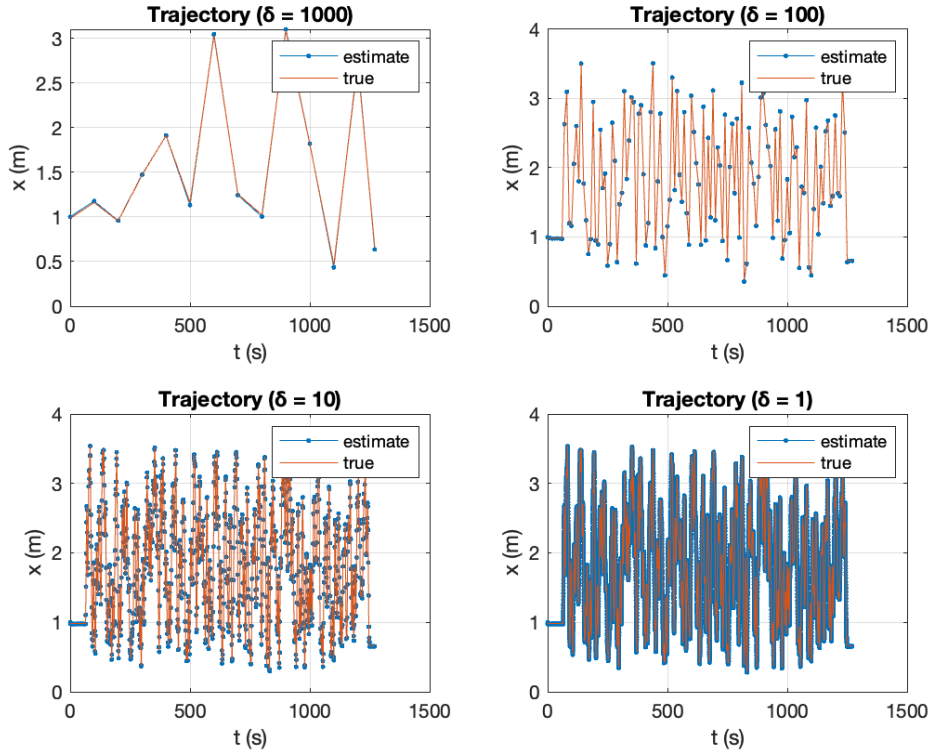


Figure 2: Estimated trajectory vs. ground truth for different values of δ .

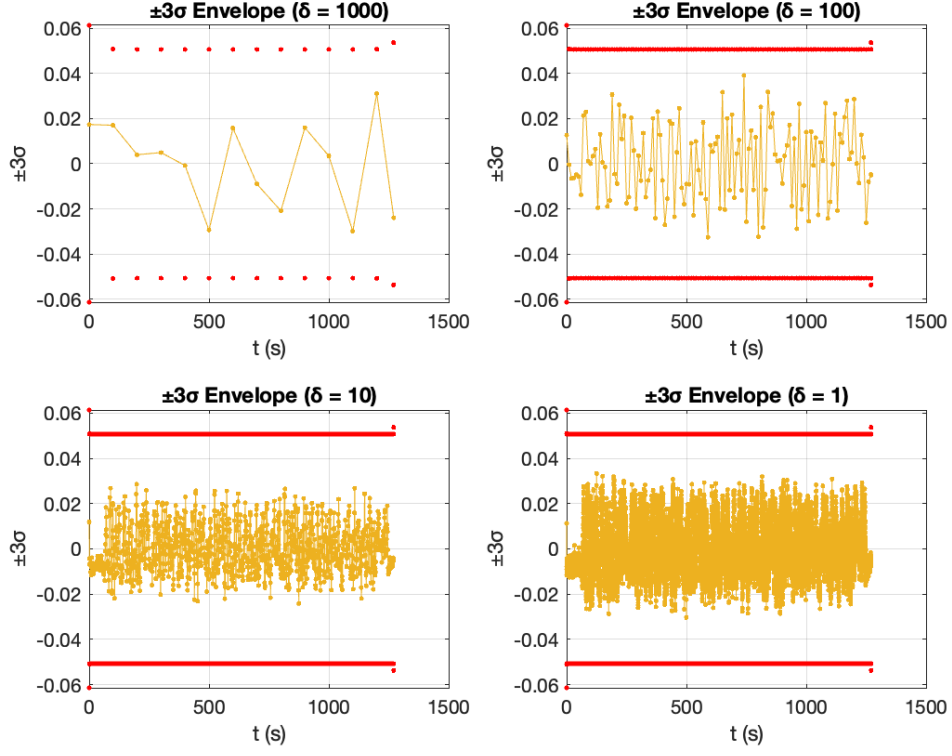


Figure 3: Predicted uncertainty envelope $\pm 3\sigma_{x_k}$ for each downsampling factor δ .

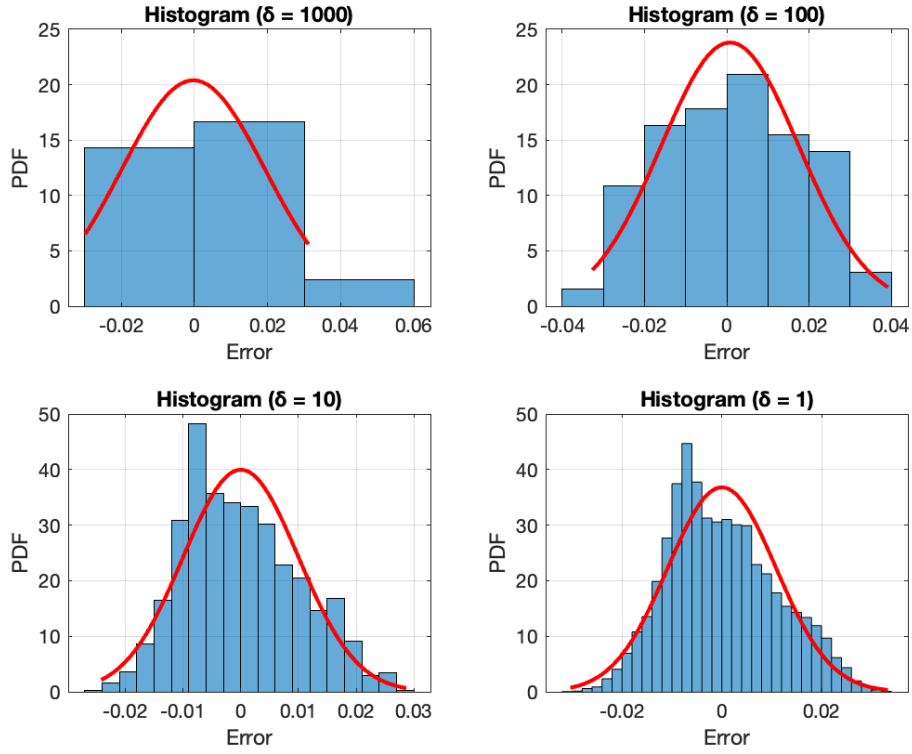


Figure 4: Error histograms with Gaussian fitting for different δ .

- tighter uncertainty bounds,
- a trajectory that better matches the ground truth,
- and a more Gaussian error distribution with smaller variance.

Thus, $\delta = 1$ yields the best estimator performance, while larger δ reduce computational cost but noticeably degrade accuracy.